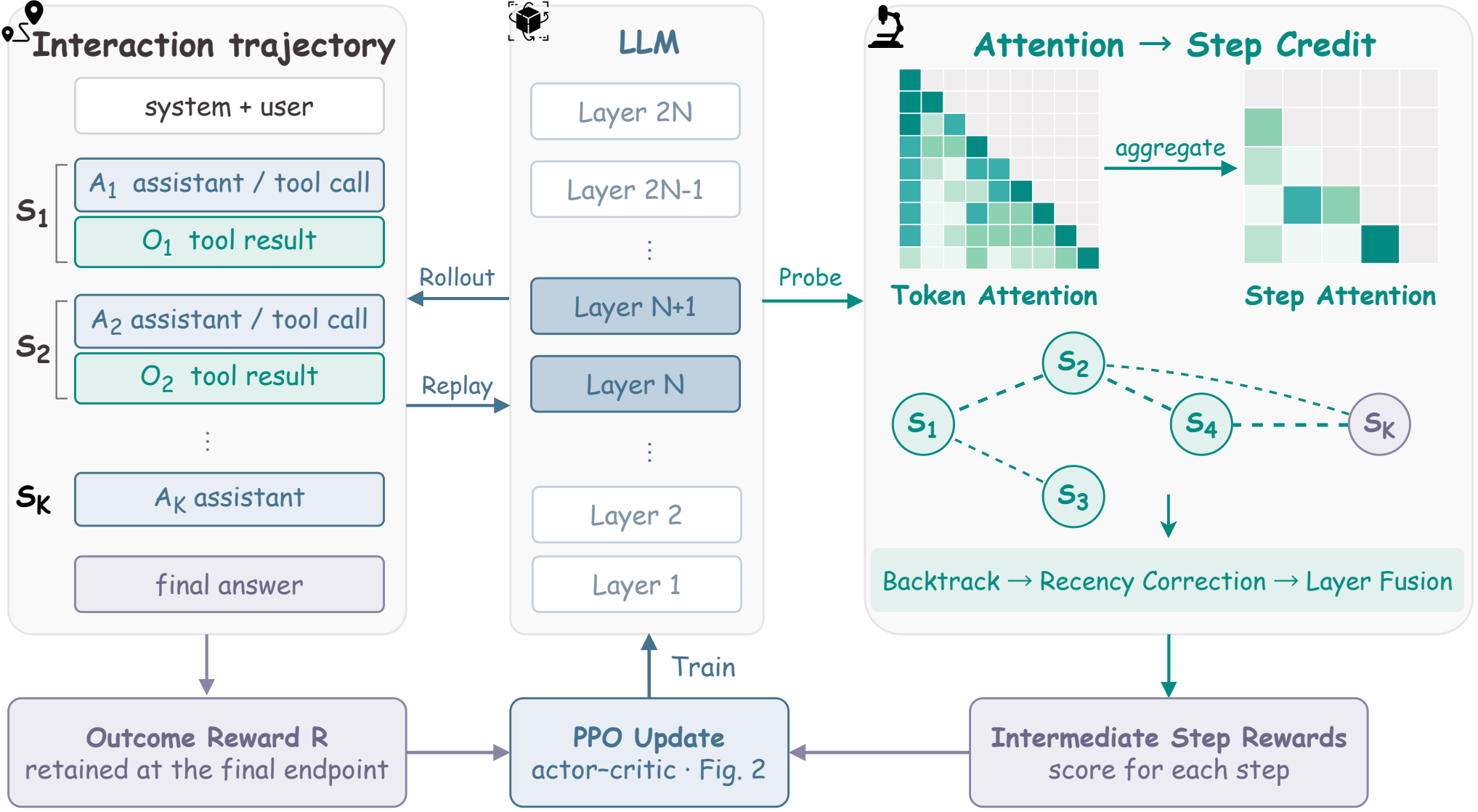

Interaction trajectory
system + user
S_1
A_1 assistant / tool call
O_1 tool result
S_2
A_2 assistant / tool call
O_2 tool result
S_K
A_K assistant
final answer
Rollout
Replay
LLM
Layer 2N
Layer 2N-1
Layer N+1
Layer N
Layer 2
Layer 1
Probe
Attention → Step Credit
aggregate
Token Attention
Step Attention
S_1
S_2
S_3
S_4
S_K
Backtrack → Recency Correction → Layer Fusion
Train
Outcome Reward R
retained at the final endpoint
PPO Update
actor-critic · Fig. 2
Intermediate Step Rewards
score for each step